Introduction and Scope

Machine learning on source code has matured alongside static analysis: both need structured representations of programs, but learning pipelines additionally need fixed feature layouts, batchable graphs, and reproducible exports [@allamanis2018survey]. COGANT addresses that gap by making the path from repository checkout to **Generalized Notation Notation (GNN) bundles** explicit, inspectable, and configurable.

Terminology: "GNN" in this manuscript

In this manuscript, **GNN** refers to the Active Inference Institute's **Generalized Notation Notation** (repository) [@friedman2024gnn], a structured notation for state-space and process models—*not* graph neural networks. COGANT translates source code into this notation, producing program-graph and state-space artifacts (state space, observation modalities, actions/policies, likelihood structure, preferences, time settings, parameterization) that downstream tools can consume. Those downstream tools include graph neural network training pipelines, which can ingest the exported tensor views of the program graph, but such neural networks are a